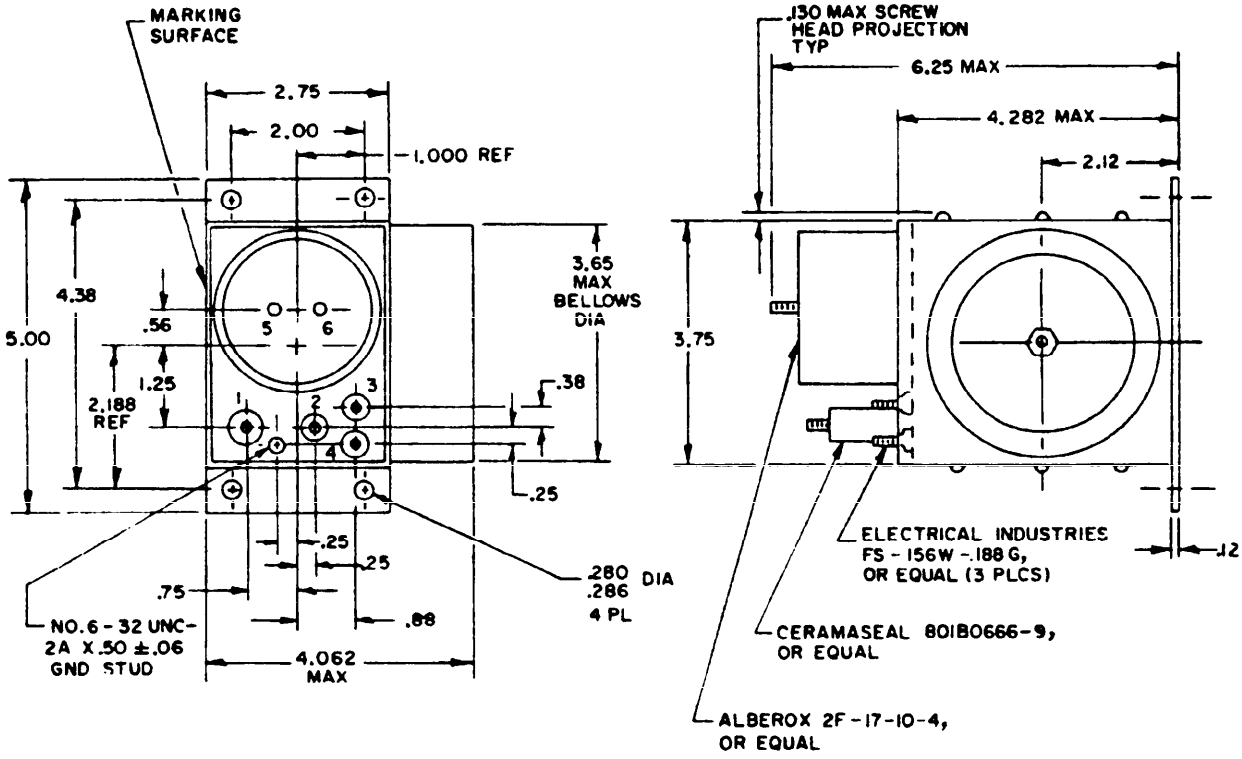
MILITARY SPECIFICATION SHEET

TRANSFORMER, PULSE, HIGH-POWER, TF4S36YY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



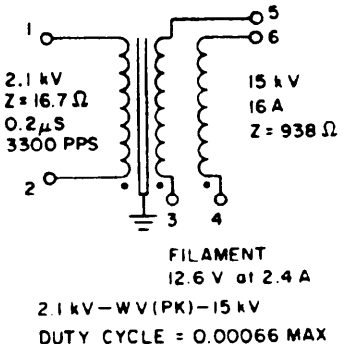
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances on all dimensions shall be ±.03 inch (.8 mm).

FIGURE 1. Dimensions and configuration.

(A) denotes changes

FSC 5950

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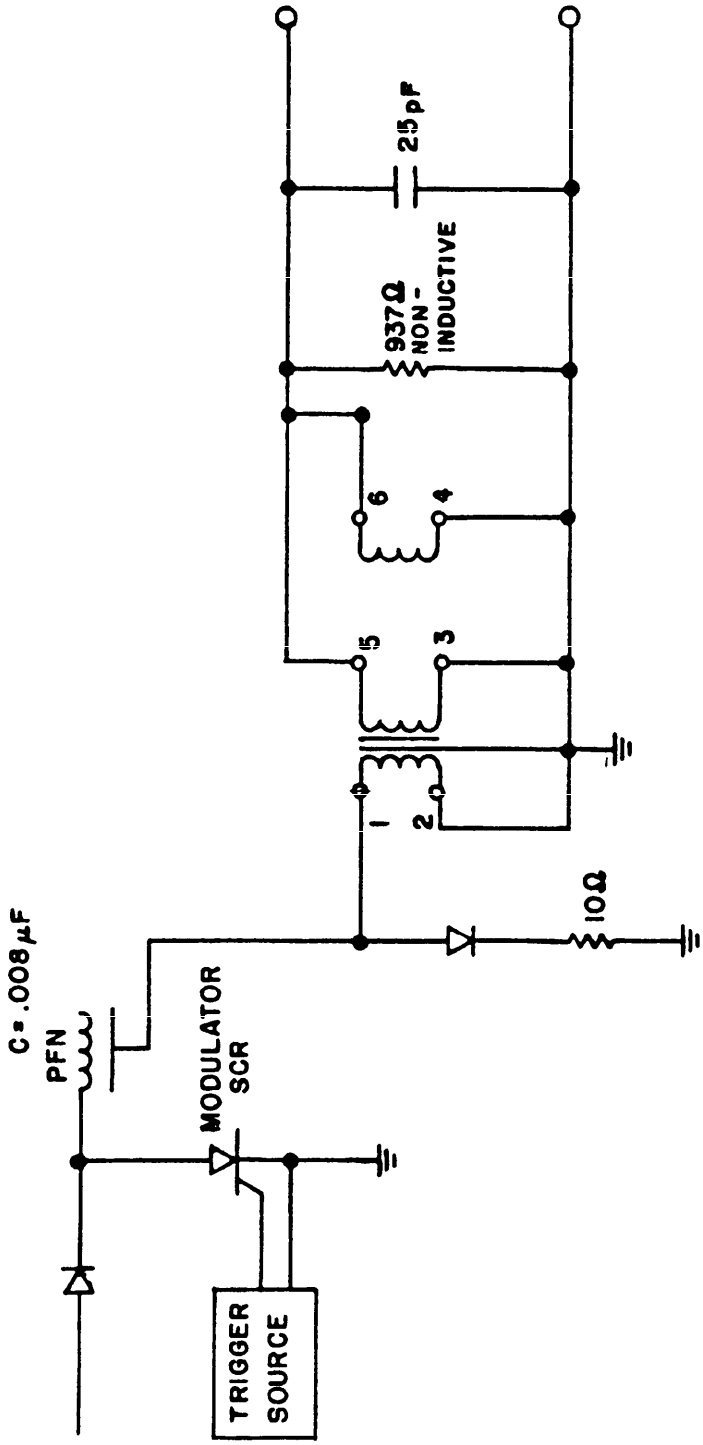


Circuit diagram and marking

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.06	1.5	.38	9.7	1.25	31.8	3.75	95.2
.12	3.0	.50	12.7	2.00	50.8	4.062	103.17
.130	3.30	.56	14.2	2.12	53.8	4.282	108.76
.25	6.4	.75	19.0	2.188	55.58	4.38	111.3
.280	7.11	.88	22.4	2.75	69.8	5.00	127.0
.286	7.26	1.000	25.40	3.65	92.7	6.25	158.8

FIGURE 1. Dimensions and configuration - Continued.

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Ⓐ FIGURE 2. Test circuit.

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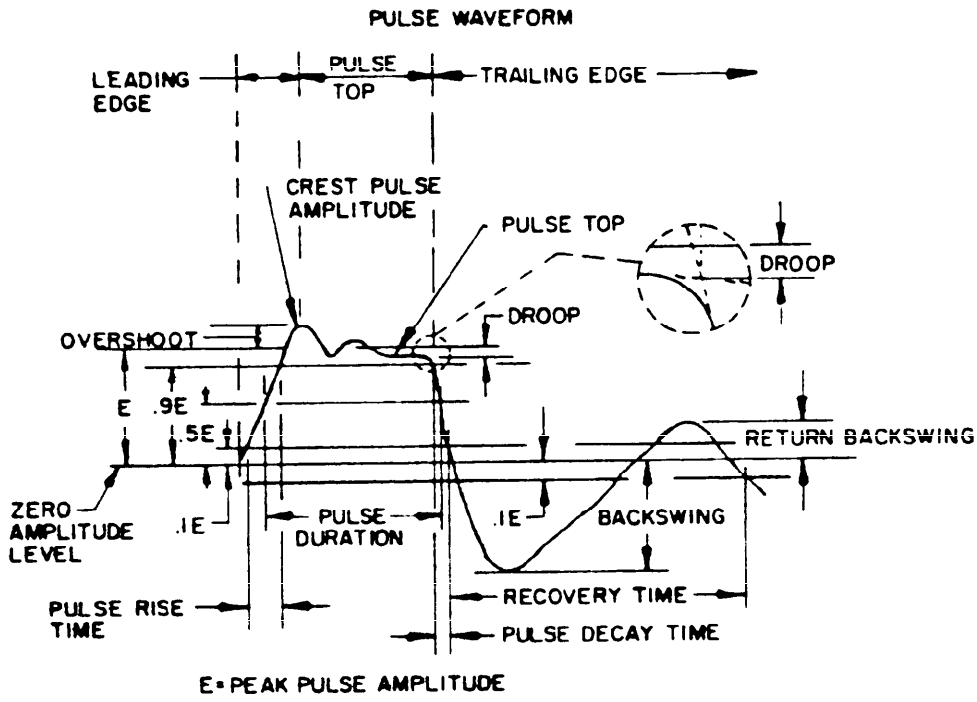


FIGURE 3. Pulse waveform.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
- Primary voltage (1-2): 2,100 volts, peak, 3,300 pulses per second.
- Primary current (1-2): 120 amperes peak.
- Secondary voltage:
  - (3-5): 15 kilovolts peak.
  - (4-6): 15 kilovolts peak.
- Secondary current:
  - (3-5): 16 amperes peak.
  - (4-6): 16 amperes peak.
- Bifilar filament current: 2.4 amperes at 12.6 volts.
- Power (average): 158 watts maximum.
- Working voltage (peak):
  - (1-2): 2.1 kilovolts.
  - (3-5): 15 kilovolts.
  - (4-6): 15 kilovolts.

Design and construction:
- Dimensions and configuration: See figure 1.
- Duty cycle: 0.00066 seconds maximum.
- Case: Metal encased.
- Terminals: Screw thread.
  - 1: Ceramaseal 801B0666-9 or equivalent.
  - 2, 3, and 4: Electrical industries FS-156W-6/32G or equivalent.
  - 5 and 6: Alberox 2F-17-10-4 or equivalent.
  - Terminal height:
    - 1: 1.531 inches nominal.
    - 2, 3, and 4: 0.281 inch nominal.
    - 5 and 6: 2.062 inches nominal.
  - Weight: 12 pounds maximum.
- Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition E.
- Torque (1, 5, and 6): 15 pound-inches.
- Torque (2, 3, 4, and GND. stud): 5 pound-inches.

Dielectric withstanding voltage (at sea level):
- (1-2) to (3-5), (4-6), and GND: 1,000 volts rms.
- (3-5) to (4-6), and GND: 1,000 volts rms.
- (4-6) to GND: 1,000 volts rms.

Induced voltage: With 15 kilovolts peak pulse, approximately 0.2 microsecond wide at 3,300 pulses per second applied to 5 connected to 6 and 3 connected to 4, the induced voltage across (1-2) shall be 25 kilovolts peak pulse, approximately 0.05 to 0.1 microsecond wide at 3,300 pulses per second. With the transformer terminals under oil and 30 kilovolts peak pulse, approximately 0.05 to 0.1 microseconds wide at 3,300 pulses per second applied to 5 connected to 6 and 3 connected to 4, the induced voltage across (1-2) shall be 25 kilovolts peak pulse, approximately 0.05 to 0.1 microseconds wide at 3,300 pulses per second.

Electrical characteristics:
- No load: With 2,100 volts peak and 3,300 pulses per second in (1-2).
  - Voltage ratio (primary winding to each secondary (2)): 1:7.5 ±1 percent.
- Rated load: With 2,100 volts peak and 3,300 pulses per second and 120 amperes peak in (1-2), the average power shall be a maximum of 158 watts at 0.2 microsecond, 3,300 pulses per second, and the voltage across (3-5) and (4-6) shall be 15 kilovolts peak.
- Impedance:
  - Source: 16.7 ohms ±5 percent.
  - Load: 938 ohms nominal.
  - Repetition rate: 3,300 pulses per second.
- DC resistance:
  - (1-2): 0.005 ohm maximum.
  - (3-5): 0.2 ohm maximum.
  - (4-6): 0.2 ohm maximum.

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Polarity: Additive, with terminals 2 to 5 and 3 to 6 connected.

Waveform parameters: The test circuit shown on figure 2 shall be used to determine the parameters.

Pulse rise time: The pulse rise time shall be a maximum of 0.15 microsecond. The method of measurement for the pulse rise time shall be as follows (see figure 3):

a. Obtain a calibrated (time and amplitude) picture of the waveform including the pulse. Delineate the pulse by excluding those portions of the waveform determined to be unwanted or nonpertinent.
b. Draw the zero axis of the pulse.
c. Find the peak pulse amplitude.
d. Draw two lines parallel to the zero axis and spaced on each side of the zero axis by 90 percent of the peak pulse amplitude and two parallel lines spaced on each side of the zero axis by 10 percent of the peak pulse amplitude. The time interval between the first point of intersection of the pulse trace and either 10 percent line and the first point of intersection of the pulse trace and either 90 percent line is the pulse rise time.

Pulse duration: The pulse duration shall be 0.2 microsecond wide. The method of measurement for the pulse duration shall be as follows (see figure 3):

a. Obtain a calibrated (time and amplitude) picture of the waveform including the pulse. Delineate the pulse by excluding those portions of the waveform determined to be unwanted or nonpertinent.
b. Draw the zero axis of the pulse.
c. Find the peak pulse amplitude.
d. Draw two lines parallel to the zero axis spaced on each side of the zero axis at 50 percent of the peak pulse amplitude. The time interval between the first and last points of intersection of the pulse trace and either line is the pulse duration.

Peak pulse amplitude: The peak pulse amplitude shall be 15 kilovolts. The method of measurement for the peak pulse amplitude shall be as follows (see figure 3):

a. Obtain a calibrated (time and amplitude) picture of the waveform including the pulse. Delineate the pulse by excluding those portions of the waveform determined to be unwanted or nonpertinent.
b. Draw the zero axis of the pulse.
c. Find the maximum departure of the pulse trace from the zero axis (regardless of polarity sign). This departure is the peak pulse amplitude.

Pulse decay time: The pulse decay time shall be a maximum of 0.1 microsecond. The method of measurement for the pulse decay time shall be as follows (see figure 3):

a. Obtain a calibrated (time and amplitude) picture of the waveform including the pulse. Delineate the pulse by excluding those portions of the waveform determined to be unwanted or nonpertinent.
b. Draw the zero axis of the pulse.
c. Find the peak pulse amplitude.
d. Draw two lines parallel to the zero axis spaced on each side of zero axis by 90 percent of the peak pulse amplitude, and two parallel lines spaced on each side of zero axis by 10 percent of the peak pulse amplitude. The time interval between the last point of the intersection of the pulse trace and either 90 percent line and the last point of intersection of the pulse trace and either 10 percent line is the pulse decay time.

Droop: The droop shall be a maximum of 10 percent (see figure 3).

Temperature rise: 59°C with 2,100 volts peak, 3,300 pulses per second, 158 watts maximum, 0.2 microsecond pulse width across (1-2) at an ambient temperature of 71°C.

Corona discharge: Applicable, test circuit 2, for winding (1-2) at 2,730 volts peak between 1 and ground.

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Salt spray (corrosion): MIL-STD-202, method 101, test condition B.

Marking location: See figure 1.

Part number: M27/218-01.

NOTE: Waveform parameters used on this specification sheet are defined as follows:

1. Pulse rise time. Pulse rise time is the interval between the instants at which the instantaneous amplitude first reaches specified lower and upper limits; namely 10 percent and 90 percent of the peak pulse amplitude.
2. Pulse duration. Pulse duration is the time interval between the first and last instants at which the instantaneous amplitude reaches 50 percent of the peak pulse amplitude.
3. Peak pulse amplitude. Peak pulse amplitude is the maximum absolute peak value of the pulse, excluding those portions considered to be unwanted or nonpertinent, such as spikes.
4. Droop. Unless otherwise specified, droop is that displacement of the peak pulse amplitude shown on figure 3. Droop is expressed in volts or as a percentage of the peak pulse amplitude.
5. Pulse decay time. The pulse decay time is the interval between the instants at which the instantaneous amplitude last reaches 90 percent, and next reaches 10 percent of the peak pulse amplitude (see figure 3).

Ⓐ Substitution data: This Military transformer is substitutible for FSCM 98675, part number GF-72-U. This in no way implies that the manufacturers part is substitutable for the military item.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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